CSCI 3060U - Phase 3: Failure Log

Digital Games Distribution Service · Arian Vares and Victor Ma · Due November 7, 2026

GitHub: https://github.com/Ary4nv/CSCI3060U-DigitalGamesProject (folder phase3-frontend-testing)

1. How to run the program and the tests

Everything is in the folder phase3-frontend-testing of the zip. It needs g++ (C++11) and bash, so it works on Linux and on the Mac.

Build and run the Front End by hand

```
cd phase3-frontend-testing
make
```

The general form of the command is (the backslash only continues the line, it can be copied as it is):

```
./distribution-system currentaccounts.txt availablegames.txt \
  gamescollection.txt transout.atf < input.txt
```

A real example that can be copied as it is, using the shared fixture files and the input of one test (it prints the terminal log and writes the daily file to out.atf):

```
./distribution-system tests/fixtures/accounts.txt \
  tests/fixtures/games.txt tests/fixtures/gamescollection.txt \
  out.atf < tests/sell/admin_can_sell/input.txt
```

The program is called distribution-system like in the handout. The four file names are command line arguments (current accounts, available games, games collection, daily transaction file). The transactions are read from standard input and everything else is printed on standard output. If the number of arguments is wrong it prints a usage message on standard error and exits with 1. The games collection file is accepted but the Front End never opens it, because the Front End only writes the daily transaction file and the Back End looks after the collection.

Run all the tests

```
cd phase3-frontend-testing
./scripts/run_all_tests.sh
```

This one command runs make first (so the newest source is always tested), runs every test, checks the daily transaction files and the terminal output, prints how many passed per group and which tests failed, and saves everything in a new folder results/run_NN. A test passes only if its terminal output and daily file match and the program exited with code 0 (a test can have an expected_code.txt with another number). If one of the helper scripts itself breaks, the run stops and exits with 2. The command exits with 0 only if all tests passed. Nothing has to be done by hand. An optional word after the command is added to the folder name (for example ./scripts/run_all_tests.sh after_fix).

Script (in scripts/)	What it does
run_all_tests.sh	does everything in one command and writes summary.txt and report.txt for the run
run_tests.sh <outdir> [filter]	handout step 5: runs the program on every test, saves the terminal output (.out), the daily file (.atf), stderr (.err) and the exit code (.code)
compare_daily.sh <outdir> [filter]	handout step 6: compares every daily file with expected_daily.txt (no expected file means no daily file may be written)

Script (in scripts/)	What it does
compare_terminal.sh <outdir> [filter]	handout step 6: compares every terminal output with expected_terminal.txt
compare_runs.sh <old run> <new run>	shows which tests were fixed, broken or new between two runs
check_formats.sh	checks the line lengths of the fixture files and expected daily files

Each test is a folder tests/<group>/<test>/ with input.txt, accounts.txt, games.txt, expected_terminal.txt and expected_daily.txt (so the input and expected files have the same test name, and the outputs of a run are saved under that name in results/run_NN/outputs/). For a test called sell_successful_sell a run saves sell_successful_sell.out, .atf, .err and .code, and .terminal.diff or .daily.diff if that check failed. The folder results/ keeps every run, which is where the failure log below comes from. There are 98 tests: the 74 from Phase 1, 21 new ones (15 for search and 6 for listaccounts, each with a description.txt) and 3 overflow tests added after run 6 (failure F18).

2. Test suite runs

We ran the whole suite 8 times. Between runs we only fixed what the failure table says. The machine each run was done on is in the table (all built with g++ -std=c++11).

#	Folder in results/	State of code and tests	Machine	Tests	Passed	Failed
1	run_01_first_run_phase2_code	Phase 2 code as it was, Phase 1 tests, new scripts using the handout command line (4 file names)	Darwin arm64	74	0	74
2	run_02_fix_four_arguments	main.cpp changed to take 4 file names	Darwin arm64	74	73	1
3	run_03_fix_deleted_username_test	wrong Phase 1 test fixed (deleted username reused)	MINGW64_NT-10.0-26200 x86_64	74	74	0
4	run_04_add_search_and_listaccounts_tests	21 new tests added (search and listaccounts), search not written yet	MINGW64_NT-10.0-26200 x86_64	95	80	15
5	run_05_add_search_transaction	SearchTransaction added	MINGW64_NT-10.0-26200 x86_64	95	95	0
6	run_06_trim_games	games.txt of 38 tests cut down to the END line (Phase 1 feedback), nothing else changed	Darwin arm64	95	95	0
7	run_07_overflow_tests_added	3 overflow tests added (addcredit, refund, sell), code not fixed yet	Darwin arm64	98	95	3
8	run_08_fix_overflow	FieldFormat fixed so amounts that are too big are refused	Darwin arm64	98	98	0

The last run is the final one: all 98 tests pass. The failed tests of every run are in the failure table, and the full list of tests that failed in run 1 is in appendix A.

3. Failure table

One row per failure. Run 2a is the first attempt at run 2. H1 and H2 are problems in our own scripts, not in the code or a test. F1 is one row because all 74 tests failed for the same reason in run 1 (every test is listed in appendix A). Run numbers are the run in which the failure was seen, the fix was checked in the next run.

#	Run	Test	What it was testing	How the output was wrong	What the error was (code or test)	What was changed to fix it
F1	1	all 74 Phase 1 tests (see appendix A)	every test, they all run the program the way the handout says: distribution-system accounts games collection daily	Terminal output was empty and no daily file was written. Standard error had the usage message (accounts_file games_file daily_transaction_file) and the exit code was 1.	Code. main.cpp (function main) only accepted 3 file names (argc != 4), but the handout command line has 4: the games collection file is the third one. The program stopped before it read any transaction. FilePaths in Session.h also had no field for the collection file.	main.cpp now checks argc != 5, stores the 3rd argument in a new gamesCollectionFile field of FilePaths (Session.h) and uses the 4th as the daily file. Usage message and header comment updated. The collection file is not opened by the Front End. The Makefile builds distribution-system. Run 2: 73 of 74 passed.
F2	2	create/ create_duplicate_deleted _this_session	create of a username that was deleted earlier in the same session	Test expected ERROR: username already exists. The program asked Enter user type (AA, FS, BS, SS): and took the next input line (logout) as the type, so it printed ERROR: invalid user type. Because logout was used up, the session never ended and no daily file was written.	Test. The test from Phase 1 assumed a deleted username cannot be created again in the same session. The client answered on Slack that a deleted username can be reused, and the Phase 2 code already followed that, so the code was right and the test was wrong. The test input also had no line for the user type.	Fixed the test only. input.txt now has FS after the username, expected_terminal.txt shows User created., expected_daily.txt has the 02 line, the 01 line and the 00 line. description.txt explains it. Code not changed. Run 3: 74 of 74 passed.
H1	2a	all 74 Phase 1 tests (first attempt at run 2, test harness problem)	running the suite again after F1 was fixed	The first attempt at run 2 gave 0 of 74 with the same usage message as run 1, even though main.cpp was already fixed.	Script. run_all_tests.sh only built the program when the file did not exist, so it ran the old build from run 1. Not a code or test error.	run_all_tests.sh now runs make every time before the tests (make does nothing if nothing changed). The attempt was redone as the real run 2 (73 of 74). The result files of the discarded attempt were not kept, so it has no run folder and is not counted in the run list.

#	Run	Test	What it was testing	How the output was wrong	What the error was (code or test)	What was changed to fix it
H2	5	all tests (test harness, not a test)	a test whose output is right but where the program exits with a wrong exit code	Found while reviewing the scripts after run 5. A review wrapper that printed the correct output and then exited with 1 still gave 95 of 95 passed.	Script. run_all_tests.sh only treated an exit code of 124 or more as a crash, and did not look at the exit status of its helper scripts.	run_all_tests.sh now fails any test whose exit code is not 0 (or the number in an optional expected_code.txt) and stops with exit code 2 if a helper script itself breaks. Checked by running the suite with a wrapper that always exits 1 (0 of 95 passed instead of 95), then the wrapper was removed. In use from run 6.
F3	4	search/ admin_hidden_from_buystandard	Buy-standard Thistledown7 searches Harbor and the non-selling admin ByteHarbor is hidden so No matches found. is printed.	Expected "No matches found." after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code. The search transaction (Lab 3 update) was not written yet. The command word was not known to TransactionFactory::create, so FrontEnd::handleCommand got no transaction and printed invalid transaction.	Added SearchTransaction.h/.cpp, SEARCH_COMMAND in Constants.h, the search texts in Messages.h and the search case in TransactionFactory::create. Run 5: passed.
F4	4	search/ admin_not_selling_hidden_from_standard	MoonlitKai searches Byte and the admin ByteHarbor is hidden because ByteHarbor has no game for sale so No matches found. is printed.	Expected "No matches found." after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F5	4	search/ admin_not_selling_visible_to_admin	Admin QuillFox9 searches Byte and ByteHarbor is shown because admins can see admins who are not selling.	Expected "Users found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F6	4	search/ admin_selling_visible_to_standard	MoonlitKai searches Quill and the admin QuillFox9 is shown because QuillFox9 has Paperclip Expedition for sale.	Expected "Users found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F7	4	search/before_login	Search typed before any login is refused with please login first and no search text prompt is shown.	Expected "ERROR: please login first" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.

#	Run	Test	What it was testing	How the output was wrong	What the error was (code or test)	What was changed to fix it
F8	4	search/case_sensitive	MoonlitKai searches glass in lower case which finds nothing and then Glass which finds Glass Orchard because search is case sensitive.	Expected "No matches found." after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F9	4	search/ deleted_user_gone	Admin QuillFox9 deletes Nimbus2 then searches Nimbus and the deleted user is not found.	Expected "No matches found." after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F10	4	search/empty_text	MoonlitKai enters a blank line as the search text which is refused with invalid search text.	Expected "ERROR: invalid search text" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F11	4	search/game_name_only	MoonlitKai searches CraterJax who sells two games but games are matched on name only so only the user line is printed.	Expected "Users found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F12	4	search/games_and_users	MoonlitKai searches Ni which matches the game Driftwood Signal Nine and the user Nimbus2 so the game section is printed before the user section.	Expected "Games found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F13	4	search/games_substring	MoonlitKai searches Drift which appears at the start of one game name and in the middle of another so both games are printed and nothing is written except the 00 line.	Expected "Games found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.

#	Run	Test	What it was testing	How the output was wrong	What the error was (code or test)	What was changed to fix it
F14	4	search/no_results	MoonlitKai searches Zebra which is in no game name and no username so No matches found. is printed.	Expected "No matches found." after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F15	4	search/sees_new_listing	MoonlitKai sells Hollow Paper Kite then searches Kite and the new listing is found in the same session.	Expected "Games found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F16	4	search/ sellstandard_can_search	Sell-standard CraterJax is allowed to search and finds Glass Orchard by searching Orchard.	Expected "Games found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F17	4	search/users_substring	MoonlitKai searches Lynx which is only part of the username ShadowLynxRider so one user line is printed and no games.	Expected "Users found:" after the search command but the program printed ERROR: invalid transaction (and then the lines typed after search were also read as commands). Daily file was correct.	Code, same cause as F3 (search not written yet).	Same fix as F3. Run 5: passed.
F18	7	addcredit/ amount_overflow, refund/amount_overflow, sell/price_overflow (3 new tests)	amounts and prices that are much too big to be a number (for example 184467440737095516.17)	addcredit: the amount was accepted (no error), the program went on to ask Enter username: and the rest of the input was read as the wrong prompts. refund: Refund complete. and a 05 line with 000000.01. sell: Game put up for sale. and a 03 line with price 000.01. Expected invalid amount / invalid price and only the 00 line.	Code. FieldFormat.cpp, functions parseMoney, parseMoneyField and digitsToNumber: the dollar digits were turned into a number one digit at a time with no limit, so a very long number overflowed the Cents type (long long) and wrapped around to a small number. None of the 95 tests had an amount that long. Found by a code review of the finished program (done with an AI tool) after run 6.	Added MAX_DOLLAR_DIGITS (12) and a function dollarPartFits in FieldFormat.cpp. parseMoney and parseMoneyField return false if the dollar part has more than 12 digits (leading zeros are not counted), so the transaction prints invalid amount / invalid price. The 3 tests were added first (run 7: 95 of 98), then the fix (run 8: 98 of 98).

After the changes in F1 to F17 run 5 passed 95 of 95 tests. Run 6 (after the Phase 1 feedback change to games.txt, no code change) passed 95 of 95 again on the Mac. After that a code review of the finished program found F18, which the 95 tests did not catch, so 3 tests were added (run 7: 95 of 98, the 3 new ones fail) and the code was fixed (run 8: 98 of 98). There are no failures left. The listaccounts tests added in run 4 passed the first time they were run, so they are not in the table.

4. Notes for the TA

- Phase 1 feedback: The Phase 1 feedback had three points. (1) Screenshots of the AI interactions were missing: the Phase 3 AI histories come with screenshots. (2) Requirements problems were missing from the project plan: we added requirements_problems.md and a new section 8b in phase1-frontend-tests/test_plan.md. (3) Are the games list and accounts needed in every test case: the 38 tests in login, logout, create, delete, general and addcredit never use games, so their games.txt now only has the END line (see run 6, 95 of 95).
- search is the Lab 3 update. The handout does not give the exact wording. The client answered on Slack (Nov 2) that the command word, prompt, matching and output format are up to us, that every account type can use it, that it is left out of the daily transaction file and that our reading of the admin rule is right. Our choices, also written in the README and in the tests: command search, prompt Enter search text:, case sensitive, text can be anywhere in the name, games matched on game name only, users on username, all account types may use it, nothing is written to the daily file, blank text gives ERROR: invalid search text, and an admin who is not selling a game is hidden from users who are not admins. Output is the heading Games found: with lines like name | seller | price, then the heading Users found: with lines like username | type (no credit). A heading is only printed if that part has matches, and No matches found. is printed when nothing matches. All the wording is in Messages.h and Constants.h.
- listaccounts is our name for the admin only transaction that shows all active accounts (Lab 2 update). It prints username | type | credit and is not written to the daily file. The client said on Slack (Nov 2) that there is no required name, so we kept it.
- Games collection file: the client said on Slack (Nov 2) that she cannot say without knowing our design and that reusing data we already have instead of making more files is fine, so the program takes the file name but does not open it.
- Script type: the client said on Slack that a Unix shell script or batch file is only a suggestion and that anything not obscure that the marker can use is fine. We used plain bash scripts that run the program as a black box.
- The first attempt at run 2 is row H1 in the failure table (old build tested by mistake, fixed in run_all_tests.sh).
- The test folder names are the Phase 1 names. create_duplicate_deleted_this_session still has the old name even though the test now checks that the create works.
- Runs 3, 4 and 5 were done on a Windows machine with Git Bash, runs 1, 2, 6, 7 and 8 on a Mac. On Windows use Git Bash or WSL and make sure make works in that shell (run_all_tests.sh calls make by itself). The script ends with some harmless rm/ln messages there because results/latest becomes a copied folder instead of a link; results/latest is not tracked by git.
- We do not want our Front End shared with other students, so we are not eligible for the bonus marks.

5. Who did what

Arian Vares	Victor Ma
Test scripts (run_all_tests.sh, run_tests.sh, compare_daily.sh, compare_terminal.sh, compare_runs.sh); 4 file name command line in main.cpp and Session.h (F1); Makefile and program name; runs 1, 2, 6, 7 and 8; the overflow tests and the FieldFormat fix (F18); the exit code check in run_all_tests.sh (H2); asking the client the follow up questions on Slack; this failure log and the instructions for the TA; the README files and the final zip.	Fixed the create_duplicate_deleted_this_session test (F2); wrote the 21 new tests (15 search, 6 listaccounts); wrote SearchTransaction and the Constants.h, Messages.h and TransactionFactory.cpp changes for it (F3 to F17); runs 3, 4 and 5 (on Windows with Git Bash, which also showed the scripts work there).

Each of us hands in our own AI history and fills in the group feedback form.

6. AI use

Generative AI was used in this phase. Each of us hands in our own full prompt and response history on Canvas with this submission. A copy is also kept in the ai_history/Phase 3 folder of our GitHub repository (outside the zip).

Appendix A: tests that failed in run 1 (failure F1)

Same symptom for every test: empty terminal output, no daily file, usage message on standard error, exit code 1. Where the test would not have written a daily file anyway, only the terminal check failed (shown in the last column).

#	Test	What it was testing	Checks that failed
1	addcredit/admin_adds_to_other	Admin adds credit to another user; amount is asked first, then the username	daily + terminal
2	addcredit/exactly_session_limit	1000.00 is the highest amount allowed in one session; accepted (boundary)	daily + terminal
3	addcredit/invalid_amount	Not a real number; refused without crashing	daily + terminal
4	addcredit/over_max_credit	Addition would push the account over the 999999.99 maximum; refused even though it's under the session limit	daily + terminal
5	addcredit/over_session_limit	1000.01 is over the session limit; refused	daily + terminal
6	addcredit/standard_adds_own	A standard user adds credit to their own account; only the amount is asked	daily + terminal
7	addcredit/unknown_user	Admin names a username that doesn't exist; refused	daily + terminal
8	bad_files/corrupt_accounts_file	accounts file has a bad line; login errors out with a format error	terminal
9	bad_files/corrupt_games_file	games file has a bad line; login errors after the username is accepted	terminal
10	bad_files/missing_accounts_file	accounts file doesn't exist; login errors immediately	terminal
11	bad_files/missing_games_file	games file doesn't exist; login errors after the username is accepted	terminal
12	buy/admin_can_buy	Confirms an admin account can also buy	daily + terminal
13	buy/already_owned	Buying the same game twice in one session; second attempt refused	daily + terminal
14	buy/buying_own_game	Seller tries to buy their own listing; refused	daily + terminal
15	buy/game_not_found	No game with that name exists; refused right after the name is typed	daily + terminal
16	buy/new_listing_same_session	A game listed this same session can't be bought until the next session	daily + terminal
17	buy/not_enough_credit	Buyer's credit is below the game price; refused	daily + terminal
18	buy/seller_over_max_credit	Purchase would push the seller over the maximum account credit; refused	daily + terminal

#	Test	What it was testing	Checks that failed
19	buy/successful_buy	Buy-standard user buys a game; checks both questions and the 04 daily line	daily + terminal
20	create/create_as_nonadmin	Full-standard MoonlitKai cannot create; refused before any prompt; not written	daily + terminal
21	create/create_duplicate_deleted_this_session	Nimbus2 deleted then re-created in same session still refused as already exists; only 02 and 00 written	daily + terminal
22	create/create_duplicate_existing	Existing MoonlitKai refused as username already exists; user type never asked; not written	daily + terminal
23	create/create_empty_username	Empty new username refused as invalid username; user type never asked; not written	daily + terminal
24	create/create_end_username	New username END refused as invalid username; user type never asked; not written	daily + terminal
25	create/create_success_15char	Exactly-15-char FifteenCharUser accepted (boundary); daily 01 then 00	daily + terminal
26	create/create_success	Admin creates NewTester FS; daily gets 01 (zero credit) then 00	daily + terminal
27	create/create_too_long	16-char SixteenCharUser1 refused as username too long; user type never asked; not written	daily + terminal
28	create/invalid_user_type	Admin create rejects an unsupported account type and writes only the logout record.	daily + terminal
29	delete/delete_as_nonadmin	Full-standard MoonlitKai cannot delete; refused before any prompt; not written	daily + terminal
30	delete/delete_empty_username	Delete with empty username refused as invalid username; not written	daily + terminal
31	delete/delete_end_username	Delete with END refused as invalid username; not written	daily + terminal
32	delete/delete_nonexistent	Delete of GhostAccount99 refused as no such user; not written	daily + terminal
33	delete/delete_self_admin	Admin deletes self QuillFox9; session ends at once; daily gets 02 then 00 with no typed logout	daily + terminal
34	delete/delete_success	Admin deletes Nimbus2; daily gets 02 with their credit then 00	daily + terminal
35	general/invalid_empty_line	Blank line as transaction while logged in refused as invalid transaction; not written	daily + terminal
36	general/invalid_unknown_word	Unknown word frobnicate while logged in refused as invalid transaction; not written	daily + terminal
37	general/invalid_wrong_case	Word Login with wrong case refused as invalid transaction (case-sensitive); not written	daily + terminal
38	general/permission_buy_as_sellstandard	Sell-standard CraterJax cannot buy; refused before buy prompts; not written	daily + terminal
39	general/permission_sell_as_buystandard	Buy-standard Thistledown7 cannot sell; refused before sell prompts; not written	daily + terminal
40	list/games_for_sale	Lists every game currently for sale with name, seller, and price	daily + terminal
41	list/nothing_for_sale	Games file has nothing listed; prints the no-games message	daily + terminal
42	list/read_only	Running list twice in a row gives identical output; nothing is written to the daily file	daily + terminal
43	list/reflects_new_listing	A game listed earlier in the session shows up in list right after	daily + terminal
44	list/stays_after_purchase	A bought game remains listed, since digital copies don't run out	daily + terminal
45	login/login_case_mismatch	Lowercase quillfox9 refused as no such user (case-sensitive); no daily file	terminal

#	Test	What it was testing	Checks that failed
46	login/login_empty_username	Empty username on login refused as invalid username; no session so no daily file	terminal
47	login/login_end_username	Username END on login refused as invalid username; no daily file	terminal
48	login/login_second_login_refused	Second login while logged in refused without asking username; only logout writes daily	daily + terminal
49	login/login_success_admin	Admin QuillFox9 can log in and log out; daily gets one 00 line	daily + terminal
50	login/login_success_standard	Full-standard MoonlitKai can log in and log out; daily gets one 00 line	daily + terminal
51	login/login_unknown_user	Unknown GhostAccount99 refused; next logout refused because never logged in; no daily file	terminal
52	login/transactions_before_login	Create, delete, logout before login each refused with please login first; only later login writes daily	daily + terminal
53	logout/logout_after_logout	Logout after a good logout refused; daily keeps only the first 00 line	daily + terminal
54	logout/logout_blocks_until_login	After logout, create refused; new login starts second session; daily reflects only the newer session, one 00 line	daily + terminal
55	logout/logout_not_logged_in	Logout with nobody logged in refused; no daily file	terminal
56	refund/buyer_over_max_credit	Refund would push the buyer over the maximum account credit; refused	daily + terminal
57	refund/invalid_amount	Checks negative, zero, and nonnumeric amounts in one session; all refused	daily + terminal
58	refund/refund_as_nonadmin	A standard account cannot refund; rejection happens before buyer or seller prompts.	daily + terminal
59	refund/same_person_twice	Same username entered as buyer and seller; refused before an amount is asked	daily + terminal
60	refund/seller_not_enough_credit	Seller's credit is below the refund amount; refused	daily + terminal
61	refund/successful_refund	Admin moves credit from seller back to buyer; checks all three questions and the 05 line	daily + terminal
62	refund/unknown_person	Buyer username doesn't exist; refused before further questions	daily + terminal
63	refund/unknown_seller	Refund rejects an unknown seller after accepting an existing buyer.	daily + terminal
64	sell/admin_can_sell	Confirms an admin account can also sell, not just standard users	daily + terminal
65	sell/invalid_price	Price is not a number; refused without crashing	daily + terminal
66	sell/name_already_listed	Game name already for sale by a different seller; refused, since uniqueness is checked globally across sellers	daily + terminal
67	sell/name_END	END is the file's end marker, so it can't be used as a game name	daily + terminal
68	sell/name_exactly_25	25-character game name accepted (boundary)	daily + terminal
69	sell/name_too_long	26-character game name refused before the price prompt	daily + terminal
70	sell/price_at_max	Price of exactly 999.99 is accepted (boundary)	daily + terminal
71	sell/price_over_max	Price of 1000.00 is over the 999.99 maximum; refused before anything is saved	daily + terminal

#	Test	What it was testing	Checks that failed
72	sell/sellstandard_can_sell	Confirms the positive case: a sell-standard account can list a game	daily + terminal
73	sell/successful_sell	Full-standard user sells a game normally; checks both questions and the 03 daily line	daily + terminal
74	sell/whole_dollar_price	Price typed with no decimal (15) is saved zero-padded as 015.00	daily + terminal